The Law of Diminishing Returns.

Diminishing returns refers to a situation where a firm is trying to expand by using more of its variable factors, but finds that the extra output they get each time they add one more variable factor to a fixed factor of production such as land, gets progressively less and less. This usually arises because the capacity of land for example, is limited in the short-run and the combination of the fixed and variable factors becomes less than optimal.

The law, with reference to land, states, "after a certain point, successive application of equal amounts of resources to a given area of land produces less than proportionate return".

If, for example, a farmer has one hectare of land (fixed factor) and produces the following bags of maize by employing more workers (variable factor).

Number of workers	Output per year	Addition to Output
1	100	100
2	210	110
3	300	90
4	250	-50

Scarcity, Choice and Opportunity Cost

The relationship between scarcity, choice and the forgone alternative is exhibited by a production possibilities curve or frontier, also known as the transformation curve, opportunity cost curve. It helps to explain the important Economic concept of opportunity cost.

Scarcity

Resources and commodities of decision makers no matter how large they may be are always limited, thus less than people's desires. As a result, everyone has some hard decision to make. In order to obtain something of value another good has to be given out.

A good is said to be an economic good when it is not free. And such a good is said to be scarce, so scarcity gives rise to price. For that matter, due to limited income, when one gets money or expects to get money, she may list down the different items, their quantities and prices that she intends to purchase. She then makes sure that the total amount to be spent is less or equal to the income. During the actual purchase, a rational consumer starts with the most pressing want, and then the less pressing are subsequently purchased. Such a list of priorities is what is referred to as the **Scale of Preference**

Choice

Choice arises out of scarcity and it simply means making the right decision. Because of the scarcity of resources, man must make a scale of preference. As noted above, a scale of preference is where items are arranged according to order of priority. Here, the most preferred or pressing need is placed in the first position then another item is place in the second position until the last item which is least pressing. When a consumer chooses the first item, then the second, he or she is said to be making choice.

Opportunity Cost

When a consumer uses her scale of preference to make choice, it implies that she is making a decision to satisfy/choose one set of wants. But this necessarily means sacrificing some other set.

It is apparent that the consumer is making choice because her resources cannot enable her to acquire all that she needs. In other words, it is because of her limited, rather scarce resources (e.g. income) that she chooses commodity A and leaves B for the meantime. By making choice therefore, she is sacrificing commodity B for A. It is this sacrifice which is referred to as Opportunity cost in economics. For instance, if a consumer purchases a house instead of 3 cars using good his limited income then the opportunity cost of a house is 3 cars. So opportunity cost is a real cost, and it arises out of scarcity and choice.

To simplify, assume that they are only two commodities, and if the society chooses more of one thing it must necessarily choose less or sacrifice something else, such as more of good X means less of good Y. The production possibility curve for any country is a graph showing the combination of two goods that can be produced using all of its scarce Economic resources in the most efficient manner, given a country's Economic development and technological progress.

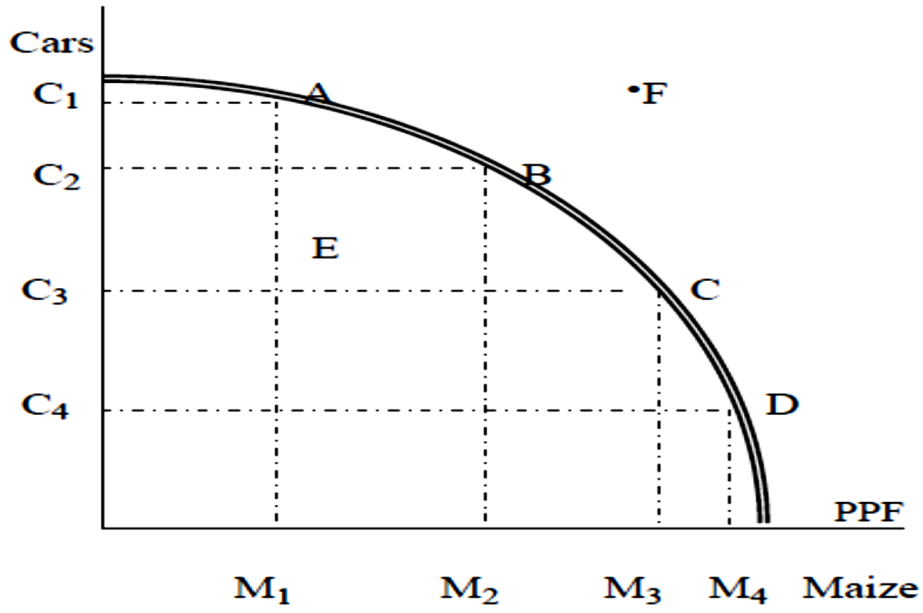
Production Possibility Curve (PPC)

A production possibility curve or production possibility frontier (PPC or PPF) is a curve that joins together the different combinations of a two goods or services that can be produced by a country or any other economic entity such as a firm when all its resources are fully and efficiently utilized.

When defining a PPF, the following assumptions are crucial:

a) Two commodities are produced e.g. cars and maize or capital goods and consumer goods
b) Technology is fixed
c) Resources are given or fixed
d) Resources are fully utilized
e) Technology is efficiently used.

Figure 1.1: The Production Possibility Curve/Frontier



Points along the PPC such as A, B, C or D show the maximum physical combination of cars and maize which can be produced when all resources are fully utilized and when technology is efficiently utilized. Along a PPC, an increase in the production of one commodity say maize can

only be done at the cost (sacrifice) of another commodity, cars (i.e. moving from point A to B on the graph). In of other words, to increase the output of maize, that of cars has to be reduced. This implies that a PPC has a negative slope especially it is concave to the origin and its slope increases as one moves down the curve.

A point inside the PPC for instance point E shows a combination of maize and cars that can be produced by the economy using less of the available resources and technology. So such a point is not desirable because it is a point of underutilization of resources. In other words, at point E, the firm A on the PPC, the economy increases the amount of cars from C_3 to C_1 and leaves the quantity of maize unaffected at M_1. Alternatively the economy could move from point E to C on the PPC and by so doing, it increases the output of maize from M_1 to M_3 leaving that of cars unaffected (i.e. at C_3). The economy can as well decide to increase the output of both maize and cars by moving from point E to any point between A and C say point B. At point B on the PPC, the output of maize and cars increases from M_1 to M_2 and C_3 to C_2 respectively.

So if an economy or firm produces inside its PPC, then it implies that either there is under employment of resources or there is use of inefficient technology or both.

It is believed that many developing countries produce inside their PPC and this is because:

a. A lot of their natural resources e.g. mineral are unexploited
b. They have poor production techniques
c. Are always embedded in political instabilities, which discourage production.

On the other hand, a point outside a PPC for instance point **F** in the above diagram, is desired by every economy or firm because it shows greater output of both maize and cars over and above the economy can potentially produce. However, such an output level can be achieved due to resource and technological constraints.

For such an output level to be achieved then there must be either one of the following or a combination of these factors:

a. Discovery of new resources e.g. new mineral deposits, land reclamation etc.
b. Discovery of new technology
c. Grants and donation

Let consider a simple example of an individual with fixed amount of land on which she produces maize and beans. Given the available land, production techniques say hoe, type of seeds used, and labour, the maximum combinations of maize and beans that she can produce is given by points along the PPF such as A, B, C, D. At the prevailing conditions, the individual can never produce beyond the PPF, however, can move to a point outside the PPF say point **F** if and only if any of the following or a combination of the following takes place.

a. He reclaims some land if she happens to live next to a forest or swamp, so as to increase on the size her land holdings: This is similar to discovery of new resources.
b. A relative or friend donates her more land.
c. She uses fertilizers or better farming techniques such as tractors, pesticides hybrid seeds etc: this is similar to adopting new production techniques mentioned before.

Finally, a movement from a point inside e.g. point E towards a point on the PPC or from a point on the PPC to a point outside the PPC shows economic growth.

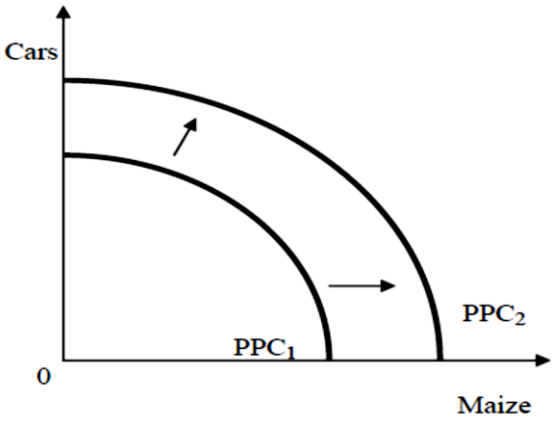
Shift in the PPC

As noted above, a movement from a point on the PPC to a point outside the PPC shows economic growth. This therefore, involves a shift in the PPC outwards to the right. However, there are two types of shifts; un-biased shift and biased shift.

Unbiased shift

Under this form, the PPC shifts parallel to the original one implying that the innovation or discovery of the new resource equally encourages the production of both commodities as shown below.

Figure 1.2: Unbiased shift in the PPF



Such a shift can be as a result of either discovery of new resources or new technology or both as long as they equally favour production of both cars and maize.

Biased Shift:

In this case the shift in the PPC is not parallel to the original one as shown below.

Figure 1.3: Biased shift

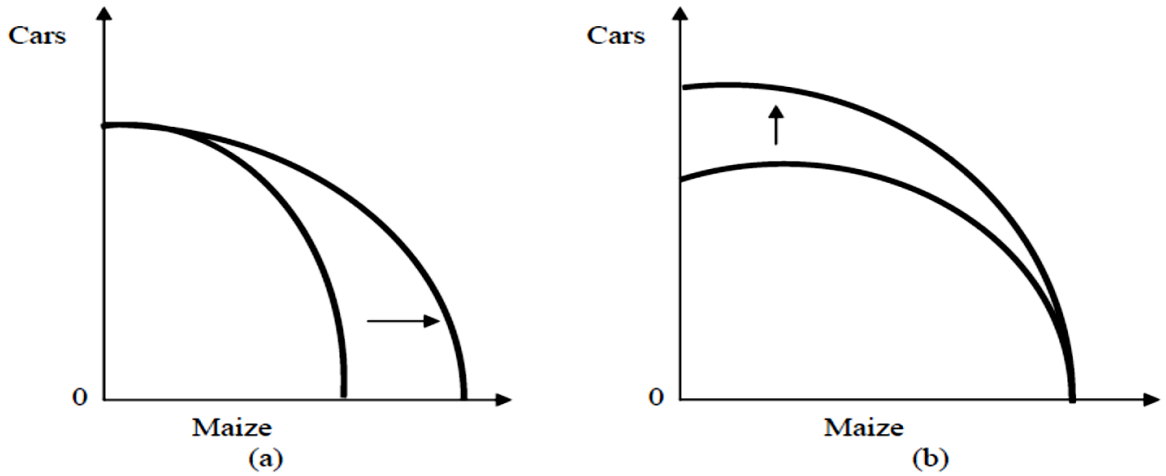


Figure (a) shows that the discovered resources or technology favours maize production and none or very little of cars. For instance a discovery of fertilizers or land reclamation would favour to a greater extent maize production than car production. On the other hand figure (b) shows that the discovered resources or technology favours production of cars none or very little of maize production.

Production possibility curve, scarcity, choice and opportunity cost

We have seen that a PPC reflects the maximum qualities of two commodities that can be produced by an economy. Also we have seen that, the slope of a PPC is negative. We however, note that this upper limit (maximum qualities) is as a result of the limited resources. So due to **scarcity** of resources, there must be a limit on the output level that an economy can produce. Also the negativity of the PPF signifies that because resources are scarce, to increase the output of one commodity, say maize than of cars because resources are scarce, to be reduced. This is as a result of the competition for the scarce factor inputs.

From the definition of the PPF, we further note that the economy has several alternative commodity combinations that it can produce at any one time. The question however, is at what particular point on the PPF, should the economy produce? In other words, should it be commodity combinations A or B or C? (**Refer to figure 1.1**). In other decision of a firm or economy to produce at any point on the PPC depends on the **Choice** of the society. If the society wants more cars than maize, then the economy should produce at point A. If on the other hand the society wants more maize than cars, then it should produce at point D and so forth. Choice is therefore, determined by social cost welfare of the society, and the type of the economic system,

Now if the society chooses to produce at either A or B on the PPC, it produces more of one commodity than another, if it chooses to produce at point B instead of A (i.e.M_2, of maize and C_2 of cars), it means that society will consume at point B instead of maize and C_1 C_2 less of cars. This implies that society is sacrificing C_1 C_2 less of cars for more M_1 M_2 of maize. So the **opportunity cost** of producing M_1 M_2 more of maize is C_1 C_2 of cars foregone.